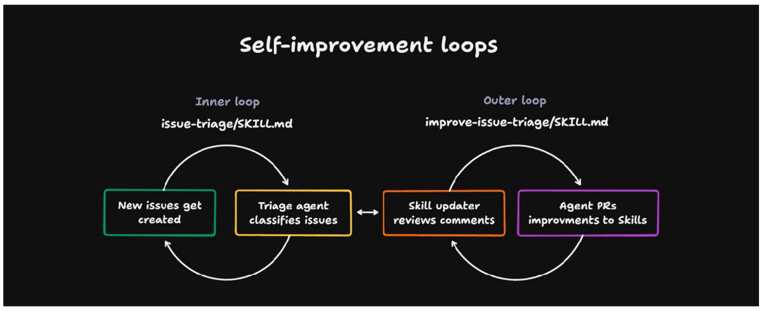
Zach Lloyd
@zachlloydtweets



How to build a self-improvement loop for your Skills

如何为你的技能打造一个自我提升的闭环

There’s been a lot of chatter about using “loops” lately to drive agents, and I think this has been accompanied by a bit of “what actually is a loop”?

最近关于用“循环”来驱动智能体的讨论很多，而随之而来的一个问题就是：“到底什么是循环呢？”

I can’t speak for everyone else using the term, but I wanted to show a practical approach using Skills and cloud agents for a particularly powerful kind of loop: a self-improvement loop.

我无法代表所有使用这一术语的人，但我想通过技能与云端智能体的结合，展示一种实用的方法，实现一种特别强大的循环——自我提升循环。

This is the idea that an agent can improve the quality of its own Skills over time from external feedback. My example is a loop that involves a human feedback step, but if you have a clear goal that doesn’t require a human, you can use the same method with an automated grader.

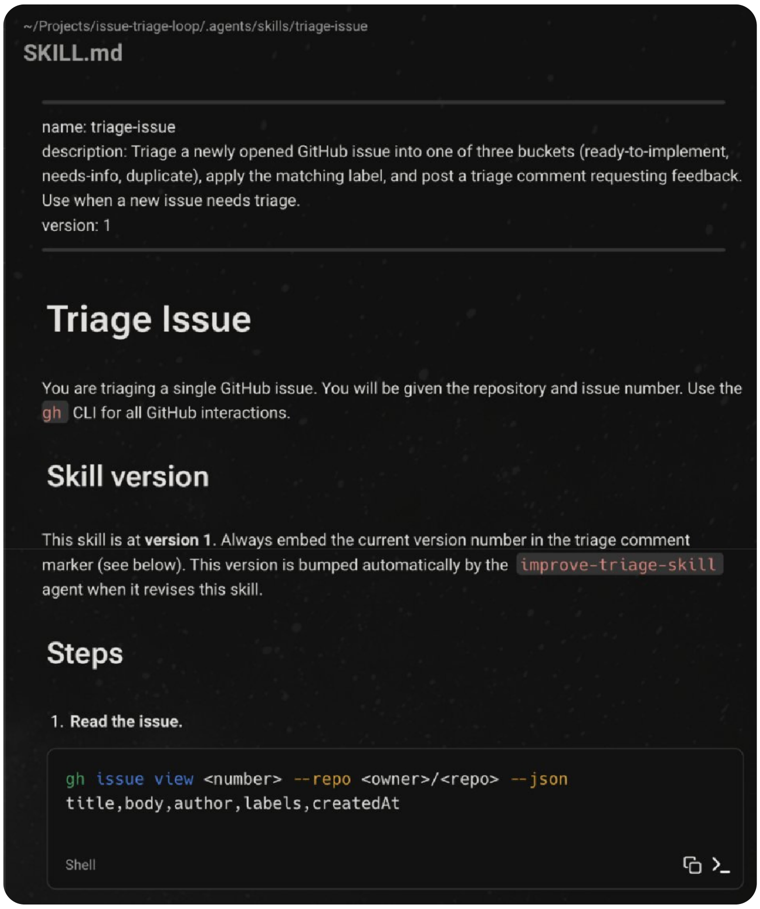
其核心思想是：智能体能够借助外部反馈，随着时间推移不断提升自身技能的质量。我的示例中包含了一个需要人类参与的反馈环节，但如果你的目标明确且无需人工介入，也可以采用同样的方法，配合自动化评分机制来完成。

To make matters concrete, say this Skill does issue triage, separating incoming issues into a few buckets: ready-to-implement, duplicate, needs-info. This would also work for a code review Skill, a bug fixing Skill, an incident response Skill, and so on.

为了更具体地说明，假设某项技能负责对收到的问题进行分类处理，将其划分为几个类别：可立即实施、重复问题、需补充信息等。这种方法同样适用于代码评审技能、缺陷修复技能、事件响应技能等等。

Here’s what a first draft of the Skill might look like:

以下是该技能的初稿可能的样子：



~/Projects/issue-triage-loop/.agents/skills/triage-issue

SKILL.md

name: triage-issue
description: Triage a newly opened GitHub issue into one of three buckets (ready-to-implement, needs-info, duplicate), apply the matching label, and post a triage comment requesting feedback. Use when a new issue needs triage.
version: 1

Triage Issue

You are triaging a single GitHub issue. You will be given the repository and issue number. Use the `gh` CLI for all GitHub interactions.

Skill version

This skill is at **version 1**. Always embed the current version number in the triage comment marker (see below). This version is bumped automatically by the `improve-triage-skill` agent when it revises this skill.

Steps

1. **Read the issue.**

```
gh issue view <number> --repo <owner>/<repo> --json
title,body,author,labels,createdAt
```

Shell

Full triage-issue Skill

完整的工单分类技能

What you need to do is set up the following loops:

您需要做的就是设置以下两个循环：

1. **An inner agent loop**: this is where you actually apply the Skill. For issue triage, you could be running it manually, or, more likely, you have an integration with your task tracker that runs the Skill whenever a new issue is filed. Interactions with the Skill are recorded somewhere: in a file, an agent trace, or an interaction in an external system like Slack or Github.

   内部代理循环：这是实际应用该技能的地方。对于工单分类，您可以手动运行它，或者更常见的是，通过与您的任务跟踪工具的集成，在每次新建工单时自动触发该技能。与该技能的交互会被记录在某个地方：例如文件中、代理追踪日志里，或是在 Slack、GitHub 等外部系统中的交互记录中。

2. **An outer agent loop**: this is an agent that runs on a schedule and observes the inner loop use of the Skill. For the issue triager, this will likely be a cloud agent that pulls records of every time the Triage agent ran. Its job is to look at all the runs of the inner agent and adjust its Skill based on the performance of those runs. Since Skills are just files, this means it should make a diff to improve Skill based

on user feedback from past runs.

外部代理循环：这是一个按计划运行的代理，用于监控内部代理对技能的使用情况。对于工单分类器而言，这通常是一个云端代理，负责收集每次“分类”代理运行的记录。它的职责是查看所有内部代理的运行记录，并根据这些运行的表现来调整其技能。由于技能本质上只是文件，这意味着它需要基于过往运行中用户反馈的内容，生成差异补丁以持续优化技能。

I'll show you how to do this in practice using Warp and Oz, our cloud agent platform, but there are lots of ways you can accomplish it. We will use Github Issues as the issue tracker.

我将通过我们的云端代理平台 Warp 和 Oz，向您展示如何在实践中实现这一过程，但其实还有许多其他方式可以达成目标。我们将以 GitHub Issues 作为工单跟踪工具。

Here is a sample repo with the Skills and GitHub workflows to follow along.

这里有一个示例仓库，其中包含了可供参考的技能和 GitHub 工作流。

Step 1: set up the inner agent loop
步骤 1：设置内部代理循环

The inner agent loop uses a Github action that runs on every new issue created.

内部代理循环使用一项 GitHub Actions 工作流，该工作流会在每个新创建的议题上运行。

triage-new-issues.yml

```yaml
name: Triage new issues

on:
  issues:
    types: [opened]

jobs:
  triage:
    runs-on: ubuntu-latest
    permissions:
      contents: read
      issues: write
    steps:
      - uses: actions/checkout@v6
      - uses: warpdotdev/oz-agent-action@v1
        id: agent
        env:
          GH_TOKEN: ${{ github.token }}
        with:
          skill: 'triage-issue'
          prompt: |
            Triage issue #${{ github.event.issue.number }} in the
            Issue title: ${{ github.event.issue.title }}
          warp_api_key: ${{ secrets.WARP_API_KEY }}
      - name: Print agent output
        run: echo "${{ steps.agent.outputs.agent_output }}"
```

Full GitHub Action

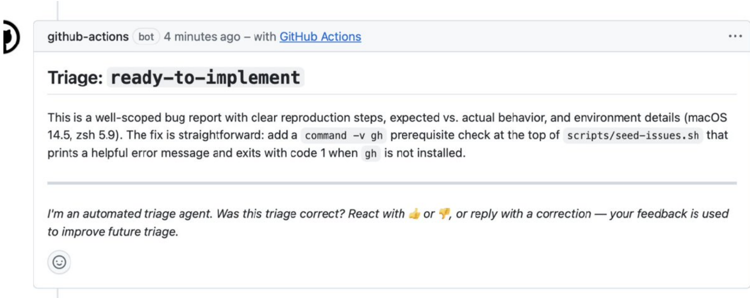
完整的 GitHub Action

The Github action invokes a cloud agent through Oz, Warp's cloud agent platform. This cloud agent syncs the repo, pulls in the issue contents from github, and tries to classify it. The code on how to set this up is in the repo linked below.

该 GitHub Action 通过 Oz——Warp 的云代理平台——调用云代理。该云代理会同步代码仓库，从 GitHub 获取问题内容，并尝试对其进行分类。相关配置的代码已收录在下方链接的仓库中。

Now when a new issue comes in, a cloud agent runs the inner loop triaging skill, and applies a label indicating that a new feature request is ready to implement.

现在，每当有新问题提交时，云端客服人员会执行内部的分类与优先级排序流程，并为其打上标签，表明该新功能需求已具备实施条件。

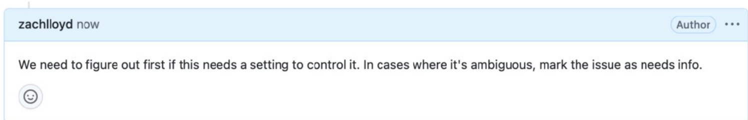


Step 2: set up the outer loop for self-improvement

步骤 2：设置自我提升的外层循环

Let’s say though that a human reviewer doesn’t agree with the agent assignment. As a person looking at the agent’s assigned labels, I switch the issue from “ready to implement” to “needs info” and add a comment on the thread as to why it was mis-categorized, e.g. because there is ambiguity on whether we should add a setting for the new feature.

不过，假设有一位人工审核员不同意该工单的分配结果。作为查看该工单分配标签的人员，我会将工单状态从“可实施”调整为“需补充信息”，并在相关讨论区添加备注，说明为何该工单被误分类，例如：由于是否应为新功能新增一项设置尚存歧义。



Here’s where the outer loop becomes interesting. The outer loop agent runs once a day and looks at all issues that have been triaged, and when it runs, it will find that I manually adjusted the label and gave a reason why.

接下来，外层循环就变得有趣了。外层循环的代理每天运行一次，会检查所有已完成分类的问题；每次运行时，它都会发现我手动修改了标签，并注明了原因。


~/Projects/issue-triage-loop/.agents/skills/improve-triage-skill

SKILL.md

name: improve-triage-skill
description: Observe how humans responded to past automated issue triage and propose improvements to the triage-issue skill via a pull request. Use on a schedule or on demand to evolve the triage skill from real feedback.

Improve Triage Skill

You are the self-improvement half of an issue-triage loop. Your job is to study how maintainers and issue authors reacted to past triage decisions made by the `triage-issue` skill (`.agents/skills/triage-issue/SKILL.md`), extract generalizable lessons, and propose edits to that skill in a pull request.

Use the `gh` CLI for all GitHub interactions. Work in the repository this skill lives in.

Step 1: Find past triage decisions

Find issues updated in the last 14 days that carry a triage comment. Triage comments always begin with the marker `<!-- oz-triage v:<N> -->`, where `<N>` is the skill version that produced them.

```
gh issue list --repo <owner>/<repo> --state all --search
"updated:>=<date-14-days-ago>" \
  --json number --limit 100
```

Shell

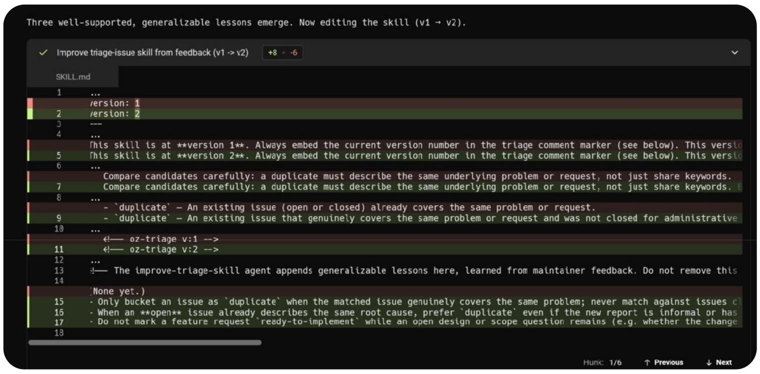
For each issue, fetch its comments and look for the marker:

[Full improve-triage-issue Skill](#)

全面改进分诊问题技能

Since the outer loop agent Skill is run through a coding agent, it will take the feedback I provided and make a diff to update the triage Skill.

由于外层循环中的代理技能是通过编码代理执行的，它将采纳我提供的反馈，并生成差异以更新分类技能。


Three well-supported, generalizable lessons emerge. Now editing the skill (v1 → v2).

Improve triage-issue skill from feedback (v1 -> v2) +8 -6

SKILL.md

```
...
version: 1
version: 2
---
...
This skill is at **version 1**. Always embed the current version number in the triage comment marker (see below). This versi
This skill is at **version 2**. Always embed the current version number in the triage comment marker (see below). This versi
...
  Compare candidates carefully: a duplicate must describe the same underlying problem or request, not just share keywords.
  Compare candidates carefully: a duplicate must describe the same underlying problem or request, not just share keywords.
...
  - `duplicate` — An existing issue (open or closed) already covers the same problem or request.
  - `duplicate` — An existing issue that genuinely covers the same problem or request and was not closed for administrative
...
  <!-- oz-triage v:1 -->
  <!-- oz-triage v:2 -->
...
<!-- The improve-triage-skill agent appends generalizable lessons here, learned from maintainer feedback. Do not remove this

(None yet.)
- Only bucket an issue as `duplicate` when the matched issue genuinely covers the same problem; never match against issues c
- When an **open** issue already describes the same root cause, prefer `duplicate` even if the new report is informal or has
- Do not mark a feature request `ready-to-implement` while an open design or scope question remains (e.g. whether the change
```

Hunk: 1/6 ↑ Previous ↓ Next

Once that diff merges, it feeds back into Skill that drives the inner loop agent, and the next time the agent runs the Skill should work better.

一旦该差异合并，它就会回流到驱动内层循环代理的技能中，下一次代理运行时，该技能应当表现得更好。

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framework behind it for others to adopt. Early version here.

很想知道这对大家是否有帮助。我们利用自我改进循环来管理 Warp 的开源代码库，并将其中的框架提取出来，供他人借鉴和采用。早期版本在此。

11:39 PM · Jun 16, 2026 · 73.6K Views

22 112 894 2.2K 2.2 千

Post your reply

guido @guidobuilds · 16h

I'm working on something similar but for skills used by coding agents running on users computers. The main problem is that users correct the agent, but the feedback never reach to the skill / knowledge owner, I'm building a skill that detect corrections and send it through an mcp

我正在开发一个类似的东西，不过针对的是运行在用户本地计算机上的编码智能体所使用的技能。目前的主要问题在于：用户会对智能体的输出进行修正，但这些反馈却无法传递给相应的技能或知识的拥有者。我正在构建一个能够识别这些修正并将其通过 MCP 发送出去的技能。

Show more

1 3 531

Zach Lloyd @zachlloydtweets · 14h

yeah the advantage of using file based skills here is that coding agents are really good at updating the skills

是的，在这里使用基于文件的技能有一个优势，那就是编码型智能体非常擅长更新这些技能。

3 377

Henry Moran @soyhenryxyz · 19h

how are you measuring if the change made a positive/negative improvement?

do you have some eval part of the system to measure?

你们是如何衡量这种变化是否带来了积极或消极的改进的呢?

你们的系统里有没有专门用于评估的部分呢?

1 3 602

Zach Lloyd @zachlloydtweets · 19h

great question. the next part of making this work well is putting an eval around it and tracking results. i'll write something up on that as well

好问题。要让这项工作顺利开展，下一步就是在其中加入评估机制，并对结果进行跟踪。我也会就此撰写一份说明。

1 3 488

Show replies

Blum @Blum_OG · 18h

moving to a self-improvement loop made my whole workflow way more effective

1 1 380

Zach Lloyd @zachlloydtweets · 17h

what's the main loop you're using

305

DC | use.fo @vibecoder_dc · 20h

This is a manager reviewing code without ever having written it. Useful? Sure. But you're optimizing a behavioral proxy from the outside.This is a manager reviewing code without ever having written it. Useful? Sure. But you're optimizing a behavioral proxy from the outside.

373

Jared Smith @woodchipdaddy · 12h

This is cool! Thanks Zach, so in this case it's particularly easy because there's already this hardened system of issue tags, and it's already part of the human's workflow to modify those. So it's elegant, i like it.

But i'm wondering if you have any examples of using this in
Show more

134

Alpha Batcher @alphabatcher · 19h
Without a doubt, information in your article can be equivalent to gold

thanks for this

245

Gabriel Tavares @gabrielgtapps · 14m
Interesting, I can see this working in a local Claude code persistent instance as well, using a local routine, memory and project & agent context.
I'll definitely try this idea out on my self reflection loop of my agents. Thanks!

4

鱼小圈 YuLoop @yyz81681981 · 4h
把"改进"落到了文件上👍。

很多self-improvement听起来像让Agent自己想更久，但这里不是。

内层跑任务，外层看反馈，然后改Skill diff。

经验不留在某次对话里，而是沉淀到下一次会被加载的harness里。

这才是真正可复利的地方。

34

DC | use.fo @vibecoder_dc · 20h
This is a manager reviewing code without ever having written it. Useful? Sure. But you're optimizing a behavioral proxy from the outside.

227

Nicholas Blanchard @corelumen · 16h
Use tugboat to keep your skills and configuration files optimized!

Implements Microsoft's SkillOpt.

syndicalt.github.io/tugboat

153

Swayam @0xsuperagent · 4h
open.substack.com/pub/superagent...

24

Niko爱学习 @ai_super_niko · 13h
By introducing external human intervention to trigger the self-evolution of the skill, perhaps we can give it a try.

90